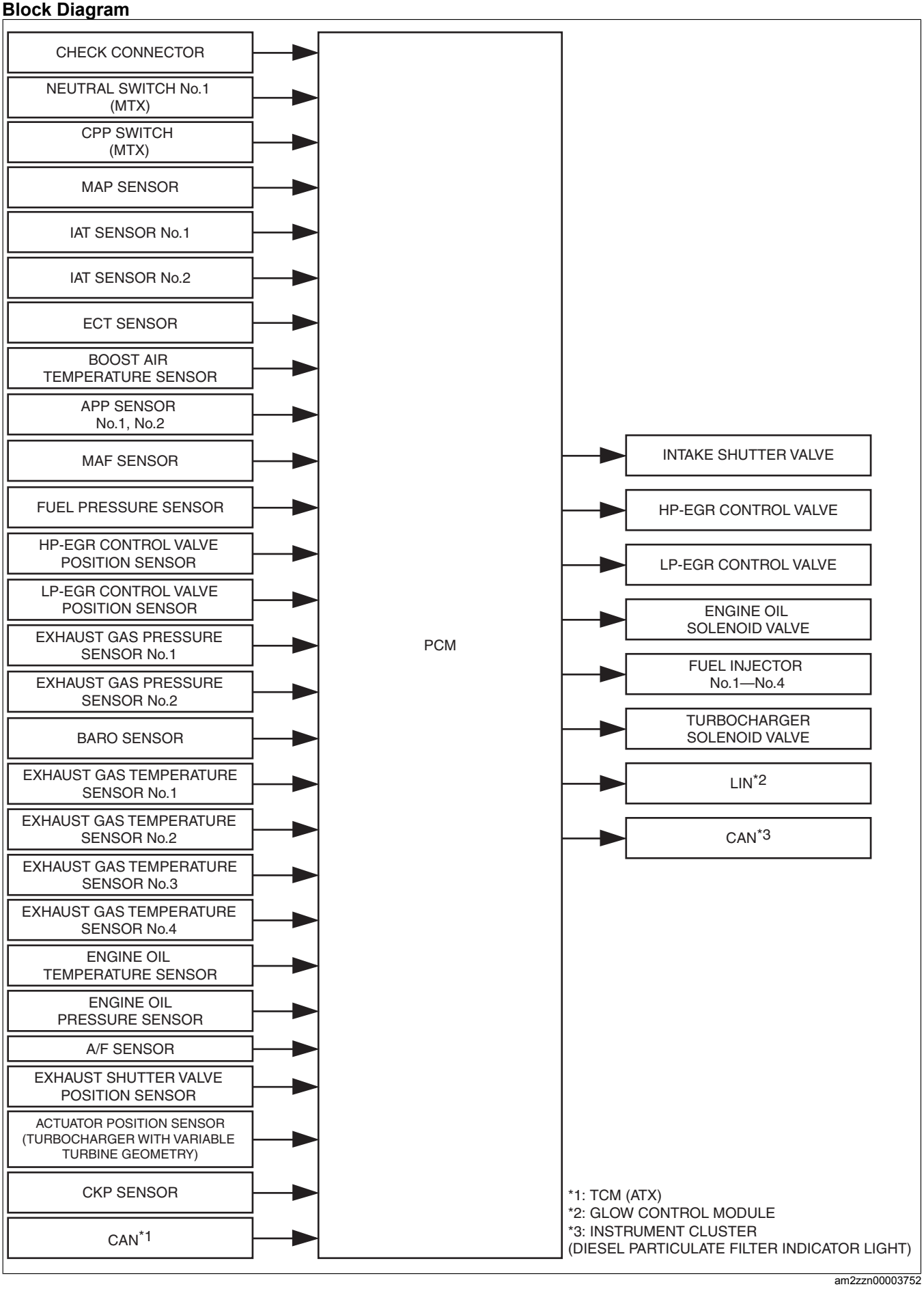
DIESEL PARTICULATE FILTER REGENERATION CONTROL [SKYACTIV-D 1.5]

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Outline

- When the amount of accumulated particulate matter (PM) in the diesel particulate filter exceeds a certain value, the PCM controls to combust and eliminate PM.
- Two methods are available to combust and eliminate PM, one is automatic DPF regeneration control which is performed by the PCM automatically, and the other is compulsory DPF regeneration control which can be forcibly performed externally.

Block Diagram



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Operation

- The PCM estimates the PM amount accumulated in the diesel particulate filter.
- Amount of accumulated PM is calculated from the difference in exhaust gas pressures and PM emission amount.
  Estimation based on difference in exhaust gas pressures
  - The amount of accumulated PM is estimated by comparing the exhaust gas pressures before and after the diesel particulate filter based on the signals from the exhaust gas pressure sensor.
- The PM emission amount is calculated from the soot generation amount and the PM combustion amount.
  Soot generation amount
  - The soot generation amount is estimated based on the engine speed, intake air amount, and fuel injection amount.

  PM combustion amount
  - The PM combustion amount is estimated based on signals from the exhaust gas temperature sensor.
- The PCM illuminates/flashes the diesel particulate filter indicator light and check engine light according to the amount of accumulated PM to promote DPF regeneration.

—: Not applicable

<table>
<tr><th>Item</th><th colspan="5">PM accumulation amount</th></tr>
<tr><td>PM accumulation amount</td><td>0—8.0 g/L {0—0.50 lb/ft^3}</td><td>8.0—10.0 g/L {0.50—0.62 lb/ft^3}</td><td>10.0—13.0 g/L {0.62—0.81 lb/ft^3}</td><td>13.0—17.0 g/L {0.81—1.06 lb/ft^3}</td><td>17.0 g/L {1.06 lb/ft^3} or more</td></tr>
<tr><td rowspan="2">Diesel particulate filter indicator light</td><td rowspan="2">—</td><td>Illuminated (Informs user)</td><td colspan="3">Flashing (Warns user)</td></tr>
<tr><td>Message related to accumulation condition in diesel particulate filter indicated in display.*1 , *2 (Informs user)</td><td colspan="3">Message related diesel particulate filter malfunction indicated in display. *2, *3 (Warns user)</td></tr>
<tr><td>Check engine light</td><td>—</td><td>—</td><td>—</td><td colspan="2">Illuminated*4 (Warns user)</td></tr>
<tr><td>Output restriction</td><td>—</td><td>—</td><td>Gas temperature restriction</td><td colspan="2">Gas temperature restriction and soot emission amount restriction</td></tr>
<tr><td>DTC recorded in PCM</td><td>—</td><td>—</td><td>P2458:00</td><td>P2463:00</td><td>P242F:00</td></tr>
<tr><td>Automatic DPF regeneration control range</td><td colspan="2">Automatic DPF regeneration control operable (If the accumulated PM amount exceeds the specified value*5, auto DPF regeneration control is implemented, and operates until the accumulated PM amount is approx. 0 g/L {0 lb/ft^3}.)</td><td colspan="3">Automatic DPF regeneration control inoperable (Automatic DPF regeneration control is disabled when amount of accumulated PM is 10.0 g/L {0.62 lb/ft^3} or more because output is restricted)</td></tr>
<tr><td>User action</td><td>—</td><td>Warm up the engine completely (engine coolant temperature 80 ° C {176 °F} or more), depress the accelerator pedal and maintain a vehicle speed of 20 km/h {12 mph} or more for 15—20 min.</td><td colspan="3">Bring vehicle to a Mazda dealer.</td></tr>
<tr><td>Mazda dealer action</td><td>—</td><td>—</td><td colspan="2">Compulsory DPF regeneration*6</td><td>Diesel particulate filter replacement</td></tr>
</table>

*1 : Turns off when it is **5.0 g/L {0.34 lb/ft^3} or less**.

*2 : For the message content, refer to the [CENTER DISPLAY]. (See CENTER DISPLAY [WITH CENTER DISPLAY].)

*3 : The diesel particulate filter indicator light flashes when the amount of accumulated PM is **10.0 g/L {0.62 lb/ft^3} or more**. This indicates that amount of accumulated PM in the diesel particulate filter is full. Therefore, to reduce the amount of accumulated PM, the fuel injection amount is reduced and exhaust gas temperature restriction (output restriction) is performed. As a result, soot generation is reduced and the PM accumulating in the diesel particulate filter is reduced. Turns off when it is **5.0 g/L {0.34 lb/ft^3} or less**.

*4 : The check engine light illuminates when the amount of accumulated PM reaches **13.0 g/L {0.81 lb/ft^3} or more**. In this case, to further restrict soot generation (output restriction), exhaust gas is induced to the cylinder by

EGR control to lower the combustion temperature. As a result, soot generation is further reduced, and the PM accumulating in the diesel particulate filter is reduced.

*5 : Figure fluctuates depending on distance travelled/conditions.

*6 : The PCM automatically selects normal mode or long mode for the compulsory DPF regeneration operation time according to the accumulated PM amount. If compulsory DPF regeneration is performed when the amount of accumulated PM is **less than 10.0 g/L {0.62 lb/ft^3}**, normal mode is performed. If compulsory DPF regeneration is performed when the amount of accumulated PM is **10.0 g/L {0.62 lb/ft^3} or more**, long mode is performed.

Automatic DPF regeneration control

- The PCM controls to combust and eliminate PM when the vehicle is driven for **10—20 min**. with all the following conditions met.
  Automatic DPF regeneration control launch conditions
  - Engine completely warmed up (engine coolant temperature: **80 °C {176 °F} or more** *1)
  - Vehicle speed: After **15 km/h {9.3 mph} or more** reached (If traffic congestion determined, after reaching **5 km/h {3 mph} or more**)
  - Accumulated PM amount: Specified value*2—**10.0*3 g/L {0.62 lb/ft^3}**
- After DPF regeneration control launches, it operates until the amount of accumulated PM amount lowers to **approx. 0 g/L {0 lb/ft^3}**.

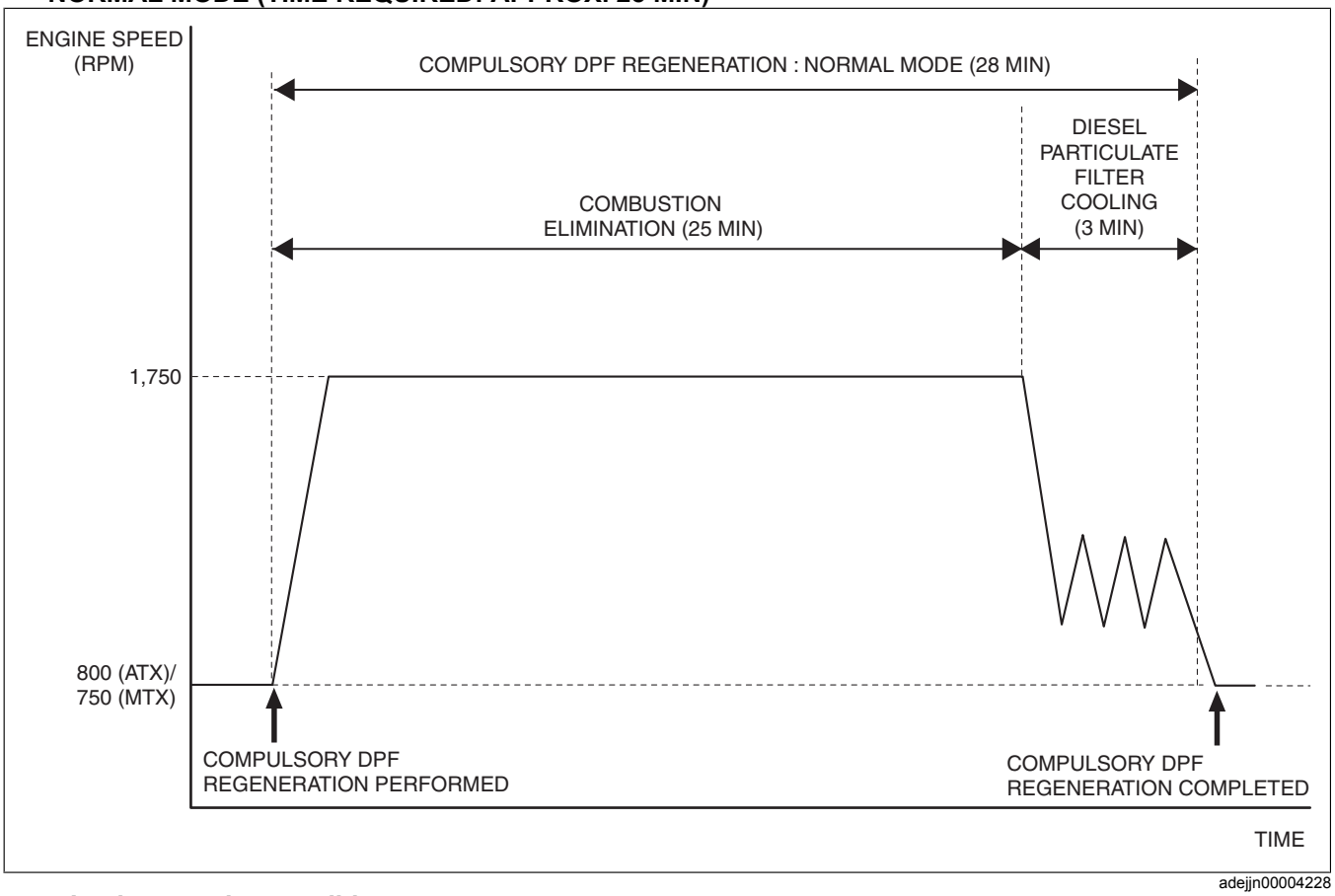
*1 : If there is little frequency of complete engine warm-up, the lowest engine coolant temperature for launch is **40 °C {104 °F} or more**.

*2 : Figure fluctuates depending on the distance traveled/conditions.

*3 : If the accumulated PM amount reaches **10.0 g/L {0.62 lb/ft^3} or more**, the output control is performed (reduction of PM exhaust amount and exhaust gas temperature), and auto DPF regeneration control is inhibited.

Compulsory DPF regeneration control

- Compulsory DPF regeneration can be performed using the Mazda Modular Diagnostic System (M-MDS).
- During compulsory DPF regeneration, idle speed is increased, and follow-up injection/post injection is performed to combust and to eliminate PM.
- The PCM automatically selects normal mode or long mode for the compulsory DPF regeneration operation time according to the accumulated PM amount.
  NORMAL MODE (TIME REQUIRED: APPROX. 28 MIN)



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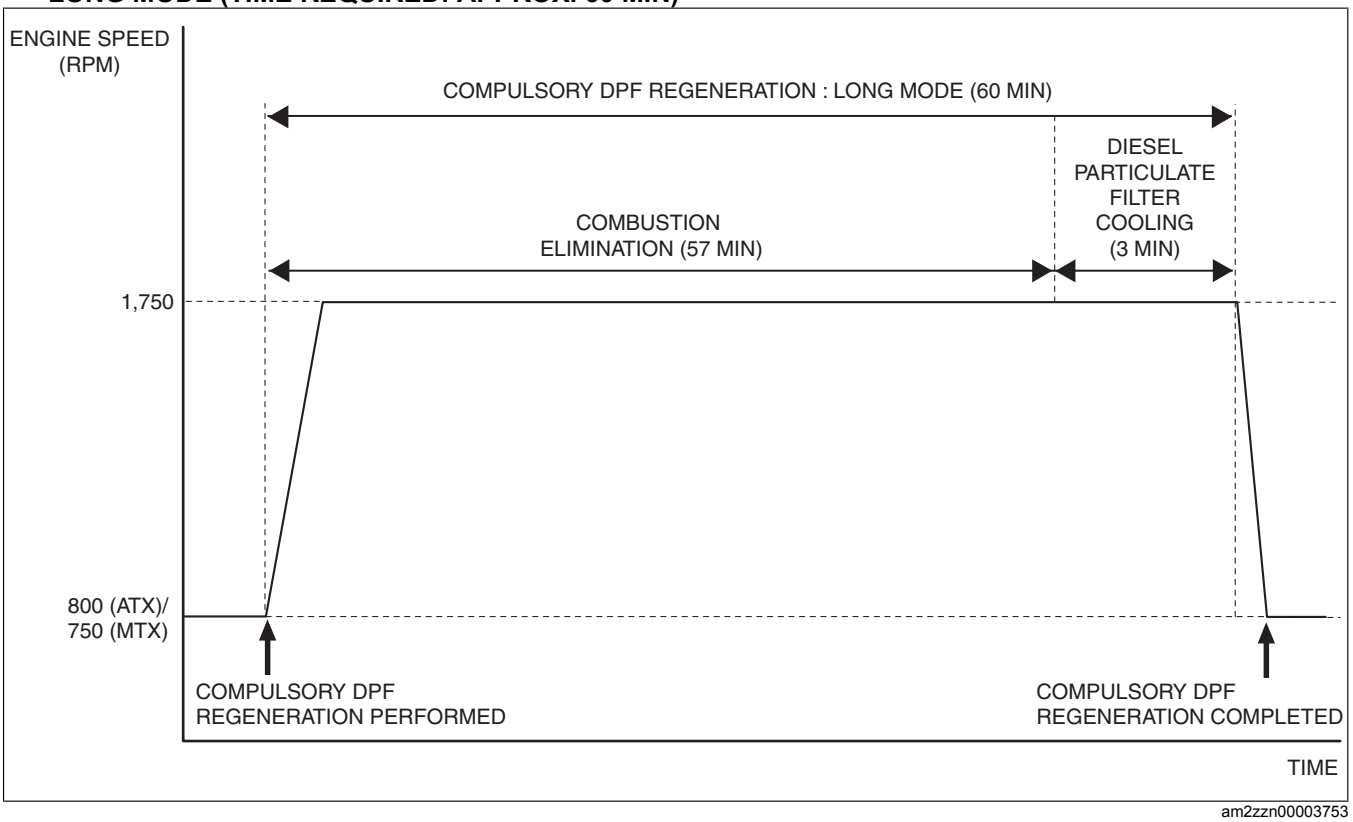
Implementation conditions

- Compulsory DPF regeneration is performed when amount of accumulated PM is **less than 10.0 g/L {0.62 lb/ft^3}**

Execution flow

1. Idle speed is maintained at **1,750 rpm** for **approx. 25 min** to eliminate PM.
2. Idle speed is decreased and maintained for **approx. 3 min** to cool the diesel particulate filter.
3. Idle speed is decreased to the normal idle speed and compulsory DPF regeneration is completed.

LONG MODE (TIME REQUIRED: APPROX. 60 MIN)



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Implementation conditions

- Compulsory DPF regeneration is performed when amount of accumulated PM is **10.0 g/L {0.62 lb/ft^3} or more**

Execution flow

1. Idle speed is maintained at **1,750 rpm** for **approx. 57 min** to eliminate PM.
2. The diesel particulate filter is cooled by inhibiting the post injection for **approx. 3 min** with the idling engine speed fixed at **1,750 rpm**.
3. Idle speed is decreased to the normal idle speed and compulsory DPF regeneration is completed.

- When the compulsory DPF regeneration procedure is performed, the PCM automatically selects normal mode or long mode according to the amount of accumulated PM. However, normal mode and long mode are not performed continuously. If compulsory DPF regeneration is performed once and the PM combustion elimination is not completed, it is necessary to perform the compulsory DPF regeneration procedure again.